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Practical - (2)

Introduction of Ecology

- Ecology is the branch of biology that studies the interaction between organisms and their environment.
- The term ecology was first introduced by Ernst Haeckel in 1866.
- It is derived from the Greek words 'oikos' (house or environment) and 'logos' (study) meaning study of home or environment.
- Ecology deals with relationships:
 - Between organisms and their physical environment (air, water, soil, climate).
 - Among organism themselves (plants, animals, microorganisms).

Level of ecological organization

1. Organism - individual living being
2. Population - Group of individuals of the same species.
3. Community - Different populations living together.
4. Ecosystem - Community interacting with the environment.
5. Biosphere - The global ecological system of Earth.

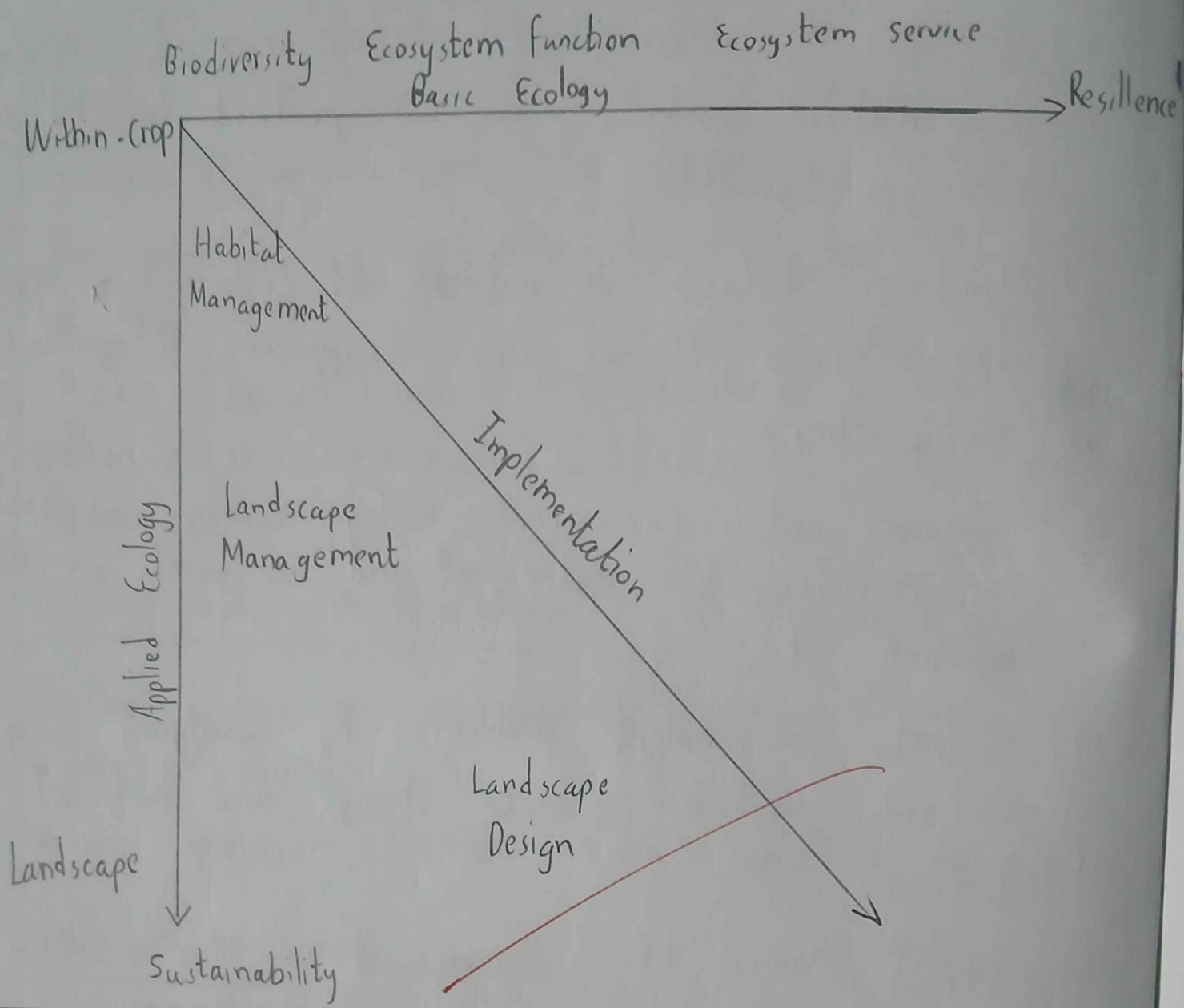
Type of Ecology

1. Autecology - Study of ~~an~~ individual species.
2. Synecology - Study of groups of organisms (communities).

Importance of Ecology

1. Protects the environment - Ecology helps us understand how nature works, so we can protect forests, oceans, rivers and wildlife.
2. Maintains balance in nature - It explains how ecosystems stay balanced. If one species disappears, it can affect many others.
3. Conserves nature resources - Ecology teaches us how to use resources like water, soil and air wisely so they do not run out.
4. Prevents environment problems - By studying ecology, we can reduce pollution, climate change, and habitat destruction.
5. Supports Human life - Humans depend on ecosystems for food, clean water, medicine and fresh air.
6. Helps in sustainable development - Ecology guides us to develop cities, industries, and agriculture without harming nature.

Ecology is important because it helps us understand nature and live in harmony with the environment. It is essential for understanding environmental balance, biodiversity, and conservation. Practical knowledge of ecology helps students observe natural ecosystems and analyze ecological relationships scientifically.



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